BOND-CDS BASIS

Global Credit Derivative and Index Research

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J.P. Morgan Securities Plc

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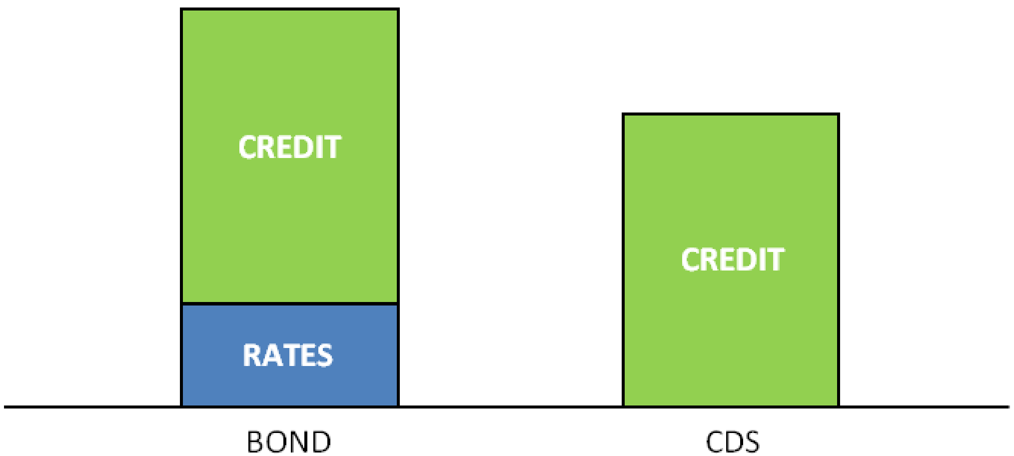
Comparing Cash Bonds and CDS

Cash Bonds and CDS

- Corporate bonds and Credit Default Swaps both reference the credit risk of the same company.
- Investors can choose to trade this credit risk through either bonds or CDS.
- However, sometimes the two different products give different prices for this credit risk.
- This is known as the **Bond-CDS Basis**.



Source: J. P. Morgan.

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Bond-CDS Basis

Measuring Bond-CDS Basis

- The Bond-CDS Basis measures the difference in the bond credit spread and the CDS spread.

Bond-CDS Basis = CDS Spread – Bond Spread

- If the basis is **positive**, CDS are trading wide to bonds: CDS cheap to cash bonds
- If the basis is **negative**, CDS are trading tight to bonds: CDS expensive to cash bonds

- Most commonly used bond spreads:
  - Z-Spread
  - Asset Swap Spread
  - Par Equivalent CDS Spread
- If the basis is **negative**:
  - Buy bonds and buy CDS protection: Negative Basis Trade / Buy Basis Package
- If the basis is **positive**:
  - Short bonds and sell CDS protection: Positive Basis Trade / Sell Basis Package

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Which bond spread to use?

Bond spreads

- To calculate the bond-CDS basis we need both a bond spread and a CDS spread.
- CDS spread is relatively straightforward – we look up the spread of a CDS contract with a maturity matched to that of the bond (interpolating if necessary).
- It's not so clear which bond spread we should use. The most common choices are:
  - **Z-Spread**. This is one of the most commonly used bond spreads with many bonds quoted on a z-spread basis. However, z-spreads are not actually traded in the market – I cannot enter a trade which pays me the z-spread.
  - **Asset swap spread**. Less commonly quoted than z-spreads, but the advantage is that you can actually receive the asset swap spread by entering an asset swap package.
- Each of these has advantages/disadvantages; we also use the **Par Equivalent CDS Spread (PECS)** (see *Bond-CDS Handbook*).
- Which risk-free rate to use?
  - Government bonds or (RFR/Libor) swaps?
  - Depends on the source of funding, choice of hedging instrument, etc.

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Trading the Basis

The Process

- Compare CDS and bond pairs and identify attractive basis opportunities.
- If basis is **negative**:
  - CDS spread is tighter than bond spread... i.e. Buying CDS protection is cheap.
  - Negative basis package: Buy bond and buy CDS protection.
- If basis is **positive**:
  - CDS spread is wider than bond spread... i.e. Buying CDS protection is expensive.
  - Positive basis package: Sell bond and sell CDS protection.
- Choose the **notionals** in each leg of the bond and CDS position to get
  - Trade sensitivity to spread movements and to default
  - Costs due to carry, upfront, funding, etc.
- Different **sizing** means different economics for basis trades. Three popular ways to trade the basis:
  - Equal notional – the most popular
  - Default weighted – no loss on principal in case of default
  - Duration weighted – not sensitive to equal moves in both spreads
- When trading the basis, be mindful about:
  - Sizing
  - Which fixed coupon to use (if alternatives are available)
  - Trade sensitivities
  - Choice of rates hedge on the bond leg (if any)
- The main risk for the trade is the mark-to-market if basis moves against the trade, especially with leverage.

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Trading the Basis – Negative Basis Trades

- When the basis is negative, the cost of protection is less than the excess credit spread we get from owning the bond.
- We can spend part of our bond spread on buying CDS protection.
- If the company defaults, we can deliver the bond into the CDS contract and receive par.
- Approximate rate of return from holding a negative basis trade: = Risk free rate + bond spread – CDS spread
  = Risk free rate – bond-CDS basis
- Trade is "risk-free" but pays more than the risk-free rate.
- The trade also benefits from any convergence in the basis.
- Approx. P&L from holding basis trade:

  ≈ (Exit Basis – Entry Basis) x Duration - Entry Basis x Days/360 – Funding Costs

- **However, trade is not "risk-free" – there are potential pitfalls with negative basis trades**.
- Very important to check **deliverability** criteria – is bond deliverable into CDS contract?

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Negative Basis Trade P&L

- For a negative basis trade, we expect to receive an annual return equal to the risk-free rate minus the bond-CDS basis.
  - e.g. if the risk free rate is 200bp and the bond-CDS basis is -50bp, the average return of the negative basis trade will be 250bp [=200bp - (-50bp)].
- This is like a bond yield – it represents the expected average return to maturity but in reality the annual payments can be structured and distributed in time differently.
- To get the whole picture, we need to consider the cost we pay to enter the negative basis package and what happens **on default**.
- **Basis Package Cost = Bond Price + CDS Upfront**
  - e.g. if the bond price is 90 and the CDS upfront for a buyer of protection is 5%, then the upfront cost of entering a negative basis trade is 95 (= 90 + 5).
- In general, investors prefer basis trades which have a basis package cost of less than 100 (assuming equal notional sizing).
  - If the basis package cost is below 100, then an immediate default benefits the investor.
  - If the cost is above 100, then an immediate default will result in a loss as the investor only receives 100 back from the CDS payout.

Exposure in Case of Default: Negative Basis Example

- Negative basis package: buy bond and buy CDS protection
- Buy bond:
  - Pay price (incl. accrued interest)
  - Receive bond coupon
  - Get bond recovery in default
- Buy CDS protection:
  - Pay/receive upfront (incl. accrued interest)
  - Pay CDS fixed coupon (and IRS rate if rates hedged)
  - Get bond loss in default (i.e., 1 - Recovery Rate)
- Altogether in default (bond price and all other variables in % terms):

  Bond notional x (Recovery + Bond coupon - Bond price) +

  CDS notional x (1 - Recovery - CDS upfront - CDS coupon)

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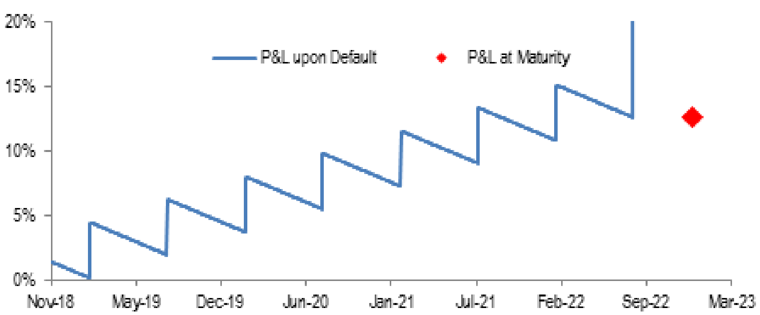
Example: Negative Basis Trade PnL and IRR on default timing

Issuer:	ABC Corp		
Bond Maturity:	01-Sep-2028	CDS Maturity:	20-Dec-2028
Coupon:	8.625% Semi-Annual	Coupon:	500bp Quarterly
Price:	61.6pt	Upfront:	34.4%
Bond Spread:	2400bp	CDS Spread:	1900bp

- **Headline Bond CDS Basis = -500bp**
- We buy bond at 61.6pts and buy CDS protection at 34.4pts upfront.
  Total package cost = 96pts (clean) and 98pts (dirty – incl. 2pt in accrued coupons)
- As the package costs below 100pt, we would be happy for default to happen straight away; we would make an instant 2pts.
- Going forward we receive annual **net** coupon of 3.625% (but pay CDS coupons for slightly longer than we receive bond coupons). If the bond is repaid at maturity, we gain 13%.
- CDS accrues coupon daily, while bonds accrue quarterly, leading to a zig-zagging PnL profile.



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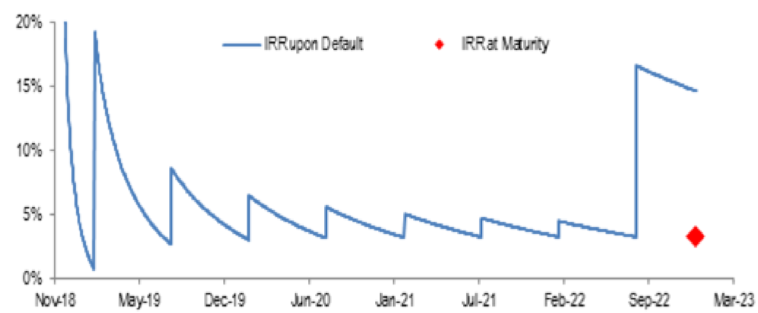

Source: J. P. Morgan.

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Sometimes negative basis trades end badly though...

SNS Bank NV

- SNS Bank subordinated debt was expropriated by the Dutch government in February 2013
  - Owners of sub debt found that they no longer owned the subordinated debt and they had received zero in return.
  - However, CDS requires a deliverable in order to calculate recovery.
    No deliverable = Recovery at 100% and no CDS payout
  - SNS bond holders could not deliver their bonds as they did not own them, Sub CDS contracts paid out very little.
- Negative Basis Trades on SNS Bank sub debt ended very badly
  - Bonds paid out zero.
  - CDS paid out zero.
  - Package was worth zero.
- Sometimes the bond-CDS basis is there for a reason!
- The new CDS definitions introduced in September 2014 are designed in a way that would avoid the SNS scenario – the so-called Asset Package Delivery feature would allow buyers of CDS protection to achieve the appropriate recovery level

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Trading the Basis – Positive Basis Trades

- When the basis is positive, the CDS spread trades wide compared to the bond spread.
- Some investors try to take advantage of this by **shorting bonds and selling CDS protection**.
- Approx. P&L from holding basis trade:

$$\approx \text{(Entry Basis – Exit Basis) x Duration + Entry Basis x Days/360 – Borrow Costs}$$

- Generally, positive basis trades are more difficult to execute than negative basis trades due to difficulties in shorting bonds.
- Even if it is not possible to execute a positive basis trade, the basis still gives you a useful signal about relative value.
- **If the basis is positive then selling CDS protection can be more attractive than buying the cash bond.**

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Why does the basis exist?

CDS-Bond Basis Drivers

- Spread reflects all the risks and costs perceived by investors: credit risk, liquidity risk, and funding/trading costs.
- Differences between CDS and bond markets can drive basis away from zero.
- On aggregate, CDS-bond basis will narrow only if CDS-bond “arbitrage” is economically sensible.
- Investors are not always able to take advantage of technical reasons via positive/negative basis trades due to funding costs or other constraints, meaning the basis can persist for some time.

Drivers	Effect on Basis
Bond issuance (in illiquid and deteriorating credit conditions)	Negative Basis
Bond issuer call options	Negative Basis
Bond repo costs	Negative Basis
Funding costs	Negative Basis
Higher CDS relative liquidity (tightening spreads)	Negative Basis
Issuance of synthetic structured products	Negative Basis
Risk on Non-deliverables	Negative Basis
Bond covenants protecting bond holders	Positive Basis
Cheapest-to-deliver option	Positive Basis
Higher CDS relative liquidity (widening spreads)	Positive Basis
Soft Credit Events	Positive Basis
Unwind of synthetic structured products	Positive Basis

Source: J.P. Morgan

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Why does the basis exist?

Reasons for Negative Basis

- **Deliverability issues**
  The bond may be issued from a different entity than the one the CDS references. If the bond entity is not guaranteed and is of worse credit quality than the CDS will trade tighter than the bond.

- **Risk of orphaning**
  Bond tenders can lead to orphaned entities, meaning that CDS contracts trade on a reference entity with no deliverable debt. This can lead to negative basis on bonds which are likely to be tendered.

- **Funding costs**
  Entering negative basis packages requires cash; if investors must borrow cash to enter basis packages they will have to pay the funding costs associated with this. This means that investors will only enter negative basis trades when the basis is negative enough to offset these costs.

- **Rating constraints / market segmentation**
  Many IG bond investors are forced to sell bonds if the company is downgraded to HY. This can cause underperformance of the bonds relative to HY in the run up to and aftermath of a downgrade.

- **Differing bond/CDS liquidity in strong markets**
  Sticky bond prices during market rallies can cause CDS to outperform.

- **Bond issuance**
  Heavy supply in bond markets can cause bonds to move wider relative to CDS.

- **Quantitate Tightening**
  Bond spreads coming under pressure when central banks start to unwind their asset purchase holdings

Source: J.P. Morgan

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Why does the basis exist?

Reasons for Positive Basis

- **Arb lists pulling single names along with index**
  Clients buy iTraxx Main protection instead of selling bonds; this pulls the single name CDS wider.
- **Cheapest to deliver option**
  The bond may not be the cheapest to deliver, meaning that the CDS will trade wider (e.g. secured bond and unsecured CDS).
- **Call features**
  A bond may be expected to be called early, meaning that the spread is tighter than the CDS to the maturity date.
- **Differing bond / CDS liquidity in weak markets**
  If markets are moving wider and bond prices are “sticky”, then they may not reflect market movements as quickly as CDS.
- **Borrow cost**
  Shorting bonds can be costly, if at all possible, and incurs a charge
- **Central bank purchases**
  Central banks can be providing a structural bid for corporate bonds as part of quantitative easing / asset purchase programmes, e.g. ECB’s CSPP.

Source: J.P. Morgan

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Online resources: Bond-CDS Basis Daily Reports

Screens for EUR and USD IG & HY

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Europe
Credit Derivatives Research
05 February 2009

Bond-CDS Basis Handbook

Measuring, trading and analysing basis trades

Basis trades exploit the different pricing of bonds and CDS on the same underlying company: by taking opposite positions in a bond and CDS, investors can profit from changes in the Bond-CDS basis. On aggregate, bond spreads are currently trading very wide to CDS spreads, generating very attractive negative basis trades (i.e. buy the bond and buy CDS protection). **This handbook reviews the definition, measurement, construction, risks and sensitivities of Bond-CDS basis trades.**

Section 1: Introduction
Section 1 introduces the concept of the Bond-CDS basis, what it encapsulates, its definition, measurement, main drivers and how it is generally traded.

Section 2: Bond and CDS: Credit Spread Measures
Section 2 tackles the measurement of the basis by analysing different spread measures for bonds (e.g. asset swap spreads, Z-spreads) and how well suited they are for comparisons with CDS spreads. We present a bond spread measure (par equivalent CDS spread or PECS) which we consider better suited for measuring the basis.

Section 3: Trading the Basis
Section 3 focuses on the trading aspects of basis trades. We start with a simple stylised basis trade example which we use to illustrate the structural features of bonds and CDS. These features affect the mechanics and economics of basis trades, and investors should be aware of all of them and their potential impact on the profitability of basis trades. This section also reviews the most popular motives for entering into basis trades: locking-in a "risk-free" spread over the life of the trade, expressing the view that the basis will revert back to zero and profiting from a default of the company. We analyse hedging ratios as well as default and spread sensitivities.

Section 4: Historical Evolution, Outlook & Current Opportunities
In Section 4 we review the historical evolution of the basis and present our outlook for 2009. This section also provides an overview of the current state of the European and US basis. We analyse the basis breakdown by sector, rating and maturity, as well as the most attractive basis trades. We outline J.P. Morgan's new daily *European Bond-CDS Basis Report*, which will highlight the best trading opportunities and the evolution of the basis by rating, maturity and sector. It complements our existing *US Corporate High Grade Basis Report* and *US Corporate High Yield Basis Report* for the US market. All of them are available on *Morgan Markets* and provide an efficient way to track the Bond-CDS basis.

In the appendices, we include a brief reminder of CDS pricing as well as a comprehensive review of bond spread measures.

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Historical IG Aggregate Bond-CDS Basis
Basis (bp).

Source: J.P. Morgan.

Bond-CDS Basis = CDS Spread – Bond Spread

CDS vs. Bond Spreads
Electricite de France
X-axis: Maturity. Y-axis: Spread (bp).

Source: J.P. Morgan. Data as of 29 January 2009. Bond spread measured as the PECS.

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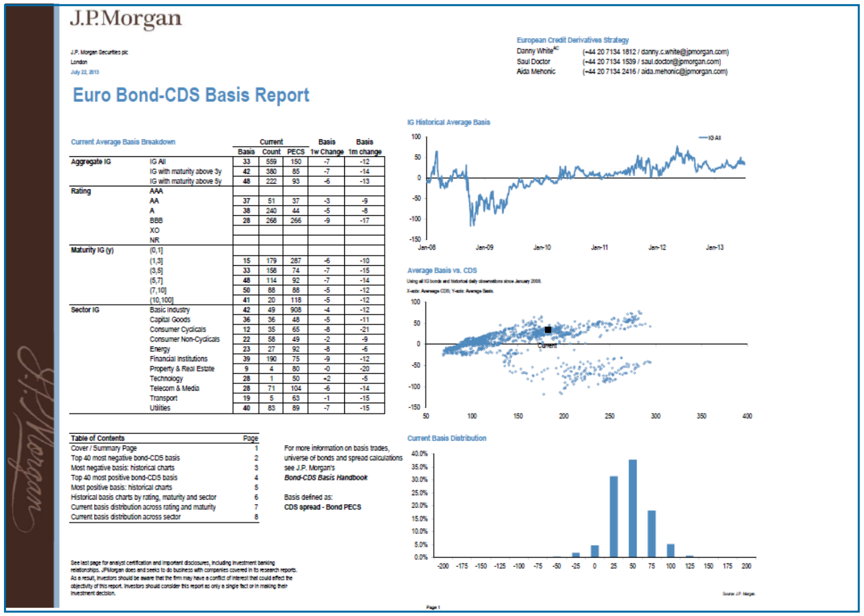

Basis reports: contain ranking of bond/CDS pairs by most positive/negative basis.

Available on Credit Derivatives Page on *JPMM* for:

- EUR IG
- EUR HY
- USD IG
- USD HY

Bond-CDS Basis Handbook

Source: J.P. Morgan

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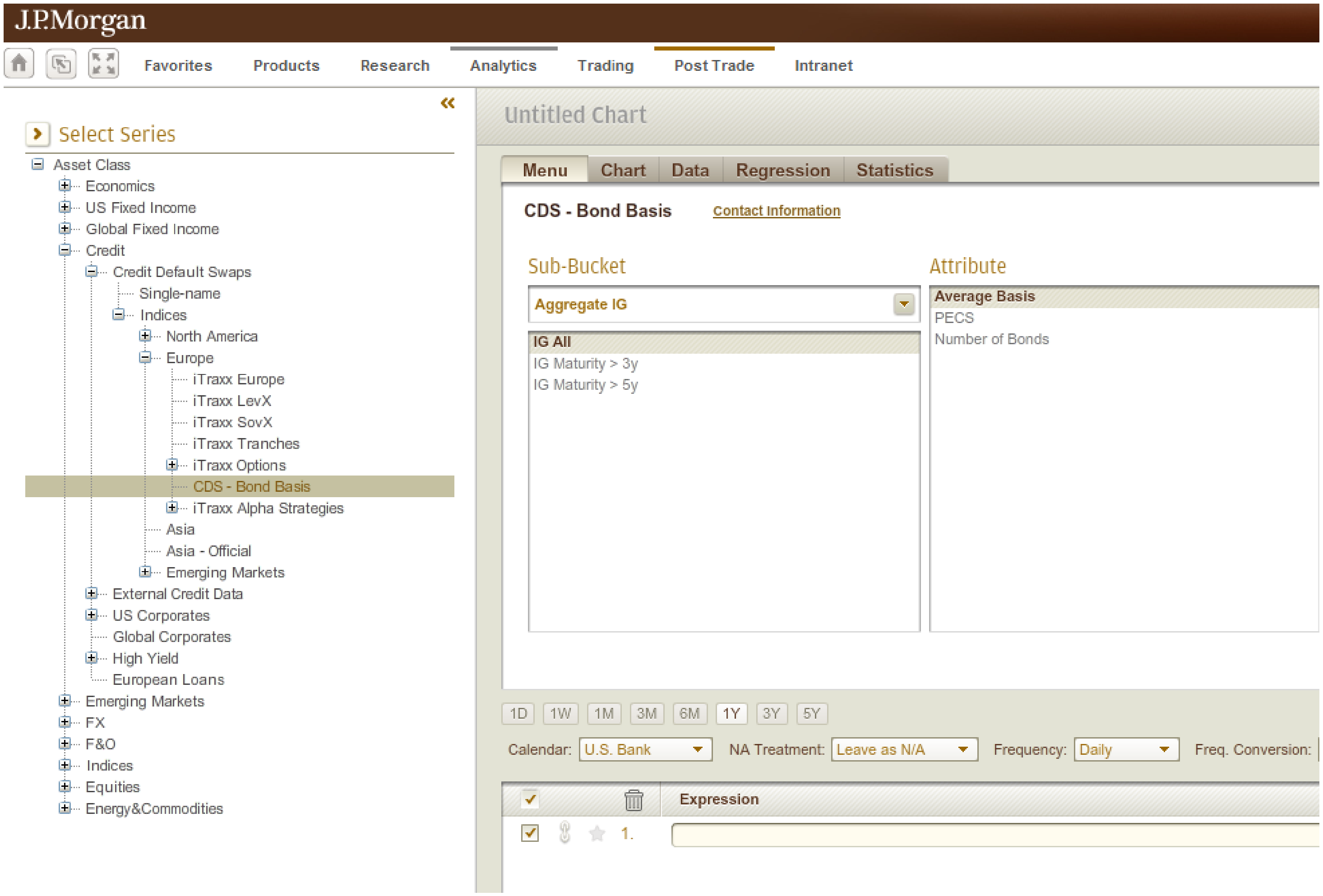
Online resources: Aggregate Basis in DataQuery

https://markets.jpmorgan.com#dataquery



Source: J.P. Morgan

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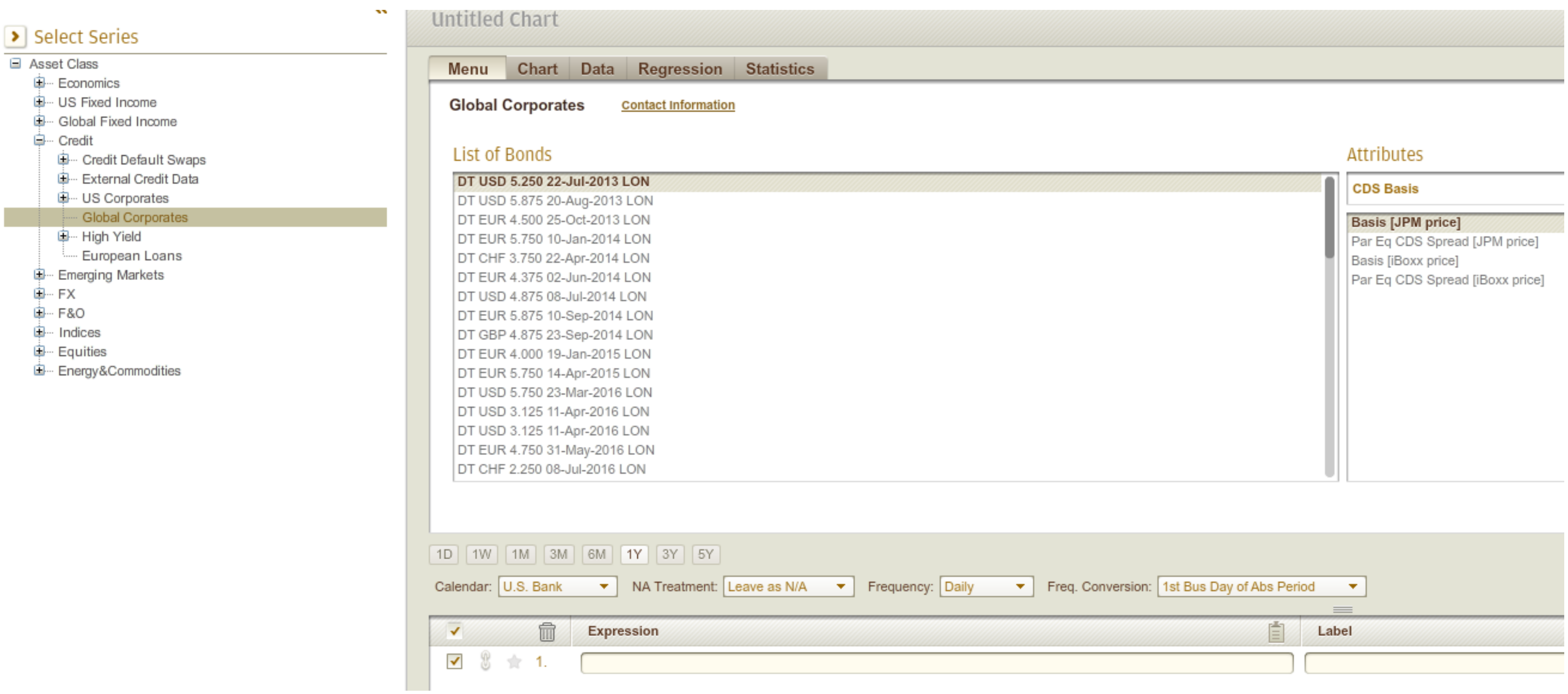
Online resources: Single Name Basis in DataQuery

https://markets.jpmorgan.com#dataquery



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Par Equivalent CDS Spread

- The Par Equivalent CDS Spread is a measure designed to calculate a bond spread using the same pricing techniques we use to price a CDS (i.e. the survival-based valuation approach). Also known as the Bond-implied CDS Spread.
- This gives a credit spread measure which is as comparable as possible with a CDS spread (i.e. consistent with the recovery rate and term structure of default probabilities priced into the CDS market).
- What do we need to calculate the PECS?
  - Bond price
  - Future bond cashflows: coupons, maturity payments, etc
  - Full swap rate curve
  - Full CDS curve
  - Assumed recovery rate
- **PECS Calculation**:
  - The CDS curve implies survival probabilities for the company at each point in the future.
  - We apply these survival probabilities to the future cashflows of the bonds, including the assumed recovery in default.
  - This gives us a "CDS-implied" price for the bond as follows:

$$\text{CDS Implied Bond Price} = \sum_{i=1}^{Maturity} (C \times PS_i + (PS_{i-1} - PS_i) \times Rec) \times DF_i + 100 \times PS_{Maturity} \times DF_{Maturity}$$

  - This CDS-implied price will be different to the bond price we see in the market; we apply a shift to the CDS curve until the CDS-implied price matches the bond price seen in the market.
  - When the two prices match, the shifted CDS spread with a maturity equal to the bond maturity is the PECS.
- Asset swap spreads and Z-spreads are useful and quick measures to calculate the basis, but they are not designed to be compared against a CDS spread. In most cases the PECS will be similar to the z-spread and asset swap spread, but there can be a large difference for distressed names and names trading with large price premiums/discounts.

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Why use PECS?

Z-spread can be misleading

- Z-spread can sometimes be misleading
  - F 8.875 2022: trades at $125
    i.e., 3.03% yield, 133bp G-sprd, and 125bp Z-sprd
  - Assume F 4.5y CDS (matures in December 2021) trades at 125bp as well
  - According to Z-sprd: Bond and CDS trade on top of each other
  - BUT in default, bond investor will lose more than CDS investor: $85 for $125 invested for bond (i.e., $68 per $100 at risk) vs. $60 if recovery rate is 40% for CDS
  - So, bond investor has more risk than CDS investor and Z-spread of an equivalent par bond (i.e., PECS) should be tighter
- How much does it cost to hedge bond with CDS?
  - In default, CDS pays $60 and bond loses $68 per $100 invested (i.e., $85 / $125)
  - $113 notional of CDS protection is needed to get paid $68 in default (60% x $113) and compensate for the bond default risk
  - Annual cost of $113 CDS protection: 125bp x $113 = 141bp
  - Therefore, bond is actually tight to its CDS, by about 16bp (i.e., 141bp – 125bp)
- How can we adjust for bond price premium/discount?
  - Approximate formula to go from Z-spread to PECS:

$$\textbf{PECS} \approx \frac{\textbf{\% loss on par}}{\textbf{\% loss on actual price}} \times \textbf{Z-spread}$$

  - In the Ford bond case: PECS is about 110bp (=125bp x 60% / 68%; since 68% = $85 / $125)
- So, in this case, F 8.875 2022 is actually tight to its CDS by approx. 15bp (i.e., 125bp – 110bp)
- Z-spread is approximately equal to PECS if bond not too far from par

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